

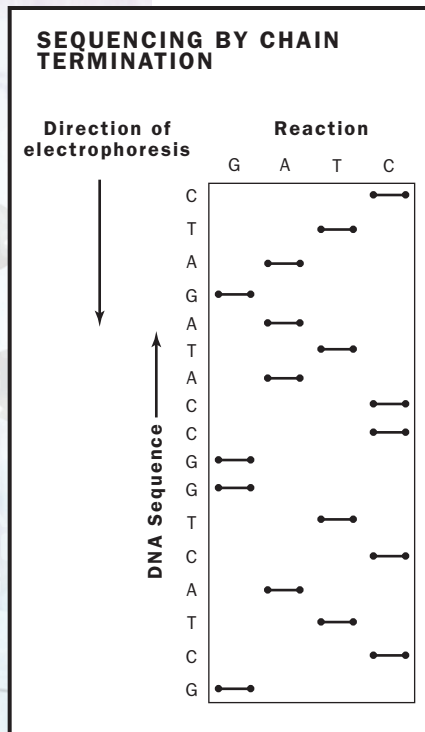


Figure 2. Schematic diagram of autoradiogram showing bands generated by the chain termination method of DNA sequencing. A band in any particular gel lane results from the incorporation of a dideoxynucleotide. DNA sequence is read from the bottom of the gel upward.

To make the DNA strands visible, a nucleotide carrying a radioactive phosphorous or sulfur atom is included in the reaction. Also included in each reaction tube is a quantity of a unique dideoxynucleotide, a modified form of a nucleotide that lacks the site at which other bases can attach during chain growth. Thus, when this nucleotide is added to the DNA chain, all further chain growth is terminated. In the “A” tube, a dideoxynucleotide form of the A base, dideoxyadenosine triphosphate, is added. In the “C” tube, dideoxycytidine triphosphate is added. Dideoxyguanosine triphosphate is added to the “G” tube, and dideoxythymidine triphosphate to the “T” tube.

When the reactions are incubated at a temperature suitable for the DNA polymerase, nucleotides that are complementary to the bases on the template are added onto the end of the attached primer. The bases A and T are complementary, as are G and C. Thus, if a T base is on the template strand, DNA polymerase will add the complementary base, A, at the corresponding position in the new, extending strand. If, by chance, the DNA polymerase adds a dideoxynucleotide, chain growth is terminated at that point.

Since dideoxynucleotides are randomly incorporated, each reaction generates a mixture of DNA molecules of variable length, but all are terminated by a dideoxynucleotide. Careful adjustment of reactant concentration will give a set of DNA molecules that terminate at each of the possible positions. The “A” tube, for instance, will contain a mixture of DNA chains, each of which ends in a different “A.”

As in Maxam-Gilbert sequencing, each reaction is loaded in its own gel lane and electrophoresal autoradiography is then used to detect the fragments. The base sequence is read directly off the X-ray film from the bottom of the gel upward, noting the lane in which each band appears (see Figure 2). The bands at the bottom of the gel represent the shortest fragments and resulted from termination events closest to the primer. Bands toward the top of the gel represent longer fragments made by termination events farthest from the primer.

Automated Sequencing with Fluorescent Dyes

As originally developed, both the Maxam-Gilbert and the chain termination methods of DNA sequencing require the use of radioisotopes to visualize the fragments generated in the reactions. However, in addition to being a health risk and presenting a disposal problem, the use of radioactivity makes automation of the DNA sequencing process difficult. Machines that could automatically read DNA sequences did not become practical until fluorescent dyes were introduced as a way to label sequencing fragments.

Two approaches are used to fluorescently label the products of a DNA sequencing reaction: the dye primer method and the dye terminator method. Both are applied only to the chain termination method. The sequencing products made by these methods are electrophoresed on an instrument that uses a laser to detect the different fragments.

In the dye primer method, fluorescent dyes are attached to the 5′ (“five prime”) end of the primer, which is the end opposite to that where nucleotides are added during chain growth. Four reactions are prepared con-

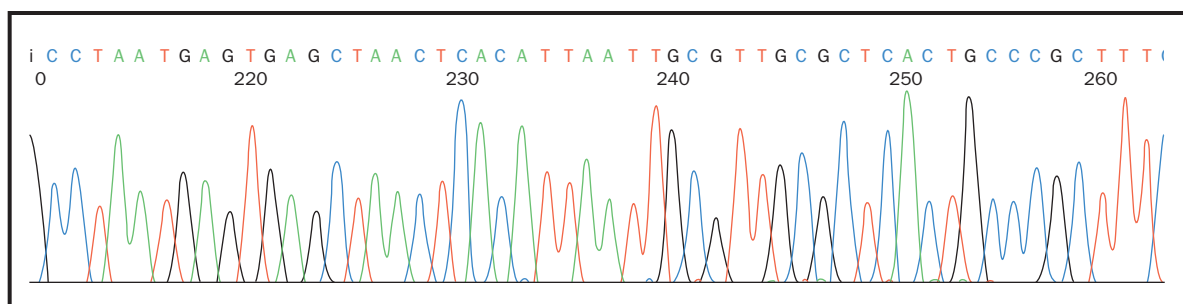


Figure 3. Chromatogram produced by an Applied Biosystems automated DNA sequencer. DNA sequence is read by a computer based on the fluorescent dye color associated with each band (represented as a peak) on the gel.

taining all the components described above, but with no radioisotopes. In the “A” reaction, the primer carries a dye at one end that fluoresces green when struck with a laser. In the “C” reaction, the primer carries a dye that fluoresces blue. The “G” reaction’s primer has a yellow dye on its end and the “T” reaction’s primer carries a red dye.

Termination events in the “A” reaction tube result in fragments with a green dye at one end and a dideoxyadenosine at the other. Termination events in the “C” reaction result in the formation of fragments having a blue dye at one end and a dideoxycytidine at the other. Similar relationships hold in the “G” and “T” reactions. When incubation is complete, the four reactions are combined and electrophoresed in a single gel lane of an automated sequencer. A laser, shining at the bottom of the gel, excites the dyes on the DNA fragments, causing them to fluoresce as they pass. The instrument’s optics system detects the fluorescent colors during electrophoresis and a computer then translates the order of colors into a base sequence (see Figure 3).

In the second approach to fluorescent DNA sequencing, the dye terminator method, dyes are attached to the dideoxynucleotides instead of to the primers. The DNA fragment in the sequencing reaction becomes dye-labeled when a dideoxynucleotide is incorporated. The dye terminator method uses a single reaction tube (rather than four) because each dideoxynucleotide is associated with a different dye. The cost and time for sequencing are therefore reduced, making this approach the preferred method used by most laboratories.

Manufacturers of DNA sequencing reagents now provide kits that contain all the components necessary for a sequencing reaction in a “master mix” format. For the dye terminator approach, a master mix can be purchased that contains DNA polymerase, the four nucleotides, buffer, magnesium, and the four dye-labeled dideoxynucleotides. The addition of **template** and primer to the master mix completes the reaction.

In addition to the widespread use of sequencing kits, other improvements have been made. Enhanced signal strength and improved sensitivity have been achieved through the development of stronger fluorescent dyes and the exploitation of heat-stable DNA polymerases that allow for repeated cycling of the sequencing reaction. SEE ALSO AUTOMATED SEQUENCER; CYCLE SEQUENCING; DNA; DNA POLYMERASES; GEL ELECTROPHORESIS; HUMAN GENOME PROJECT; NUCLEOTIDE; RESTRICTION ENZYMES.

Frank H. Stephenson and Maria Cristina Abilock

template a master copy

This automatic DNA sequencer performs “capillary electrophoresis.” This system enables the electrophoresis to complete separations faster since higher voltages are used.



Bibliography

Bloom, Mark V., et al. *Laboratory DNA Science: An Introduction to Recombinant DNA Techniques and Methods of Genome Analysis*. San Francisco: Benjamin Cummings Publishing, 1995.

Maxam, Allan M., and Walter Gilbert. “A New Method for Sequencing DNA.” *Proceedings of the National Academy of Sciences* 74 (1977): 560–564.

Sanger, Frederick, S. Nicklen, and Alan R. Coulson. “DNA Sequencing with Chain-terminating Inhibitors.” *Proceedings of the National Academy of Sciences* 74 (1977): 5463–5467.

Severe Combined Immune Deficiency

stem cells cells capable of differentiating into multiple other cell types

The development of the immune system is a very complicated process. **Stem cells** in the bone marrow continually give rise to the white blood cells responsible for producing antibodies (B-lymphocytes), recognizing and destroying foreign cells (T-lymphocytes) and performing immune surveillance for cancer and foreign cells. Immunodeficiency results when the normal complex interactions of the immune system required to do this fail, resulting in susceptibility to infections or cancer.

Types and Severity of Immunodeficiency Diseases

Disease severity can range from mild to fatal, depending upon what part of the immune system is affected. Immunodeficiency can originate in normal individuals as a consequence of chemotherapy, viral infections (such as AIDS, which is caused by the HIV virus), or as the result of other processes that prevent immune system function. When immunodeficiency occurs in this manner, it is called acquired.

In contrast, immunodeficiency can also be inherited as a genetic mutation that prevents the normal development and function of the immune system. This is called primary immunodeficiency, of which there are three subtypes: mutations that prevent the function of B-lymphocytes (antibody production), those that prevent the function of T-lymphocytes (“invader”

recognition), and those that affect both B- and T-lymphocyte production. The last group is called severe combined immunodeficiency disease (SCID). These patients make none or very few T-lymphocytes, have nonfunctional B-lymphocytes, and may or may not have a type of immune cell called natural killer cells. This combination results in the absence of a functioning immune system from the moment of birth.

SCID

SCID is a collection of rare diseases, estimated to occur once in every 80,000 live births. If left untreated it always results in fatal infections within the first two years of life. At first, to prolong life, patients were placed in sterile isolators, away from direct human contact. This gave rise to the name “bubble babies.” Currently bone marrow transplantation is available with varying degrees of success. Gene therapy to treat SCID is under development.

Although the first known description of SCID was in 1950, very little progress in understanding the genetic basis for the disease was made until the mid-1990s. SCID patients have a wide range of symptoms that make it difficult to define the number of genes involved. The development of strains of mice that exhibited SCID, either naturally or created by a laboratory technique called gene knockout, greatly enhanced the ability to study human SCID genes and permitted the development of strategies for treatment, such as bone marrow transplantation and gene therapy. Many SCID patients were able to live longer through bone marrow transplants. The study of these patients, combined with the advances from the Human Genome Project, has led to a rapid increase in our knowledge about the genetic cause of SCID. The genes involved in five different SCID diseases were confirmed in late 2001 and are described below.

SCID-X1 (XSCID, XL-SCID). This gene is currently the only known X-linked version of SCID (all other SCID forms identified are **autosomal** and recessive). SCID-X1 accounts for 46 percent of all SCID cases and exhibits a high spontaneous mutation rate. It is caused by mutations in the gene for the γ subunit of the interleukin 2 (IL-2) **cytokine** receptor. This receptor is part of a critical cytokine signal pathway required early in immune system growth and differentiation. The most famous SCID-X1 patient was David Vetter. Known as the “bubble boy,” he lived for twelve years in an isolated environment before dying from an Epstein-Barr virus infection.

ADA-SCID. This occurs in 15 percent of SCID patients. It is due to mutation in the *ADA* gene on chromosome 20. In the absence of ADA enzyme, accumulation of deoxyATP (a DNA **nucleotide**) occurs within immune cell precursors. This leads to their death through **apoptosis** within six months of birth. Unlike other forms of SCID, ADA-SCID patients can be treated through enzyme replacement therapy. Weekly injections of ADA enzyme (stabilized with polyethylene glycol) hinders toxic deoxyATP buildup, reducing apoptosis and permitting some B- and T-lymphocytes to mature. However, bone marrow transplantation provides better immunity if it succeeds.

JAK3 Deficiency. This accounts for 7 percent of SCID patients. This form of SCID maps to the Janus **kinase** 3 gene on chromosome 19. JAK3 enzyme, a tyrosine kinase, is part of the intracellular signaling pathway (JAK-STAT)

autosomal describes a chromosome other than the X and Y sex-determining chromosomes

cytokine immune system signaling molecule

nucleotide the building block of RNA or DNA

apoptosis programmed cell death

kinase an enzyme that adds a phosphate group to another molecule, usually a protein



Baby David ("bubble boy") Vetter plays in his plastic-enclosed environment. This protection helped David to survive, since he was born with severe combined immune deficiency syndrome.



that activates the genes for T-lymphocyte differentiation. To do so, it must bind to the SCID-X1 gene product.

Interleukin-7 receptor α Chain Deficiency. This is responsible for SCID in a small number of patients. This receptor is part of a larger complex that includes the SCID-X1, JAK3 proteins, and four other interleukin receptors (IL-2, IL-5, IL-9 and IL-15). Thus, at least three of the known SCID genes have been found to disrupt the same receptor complex involved in immune system development.

RAG1 and 2 Deficiencies. These can also lead to SCID. RAG1 and RAG2 are found side-by-side on chromosome 6. Mutations in either can result in SCID. They are involved in the genetic rearrangement of both the T- and B-lymphocyte receptor genes during differentiation that gives rise to the ability of the immune system to recognize foreign agents. Mutations in RAG1 or 2 also account for 50 percent of those patients with an unusual **autoimmunity** form of SCID called Omenn syndrome.

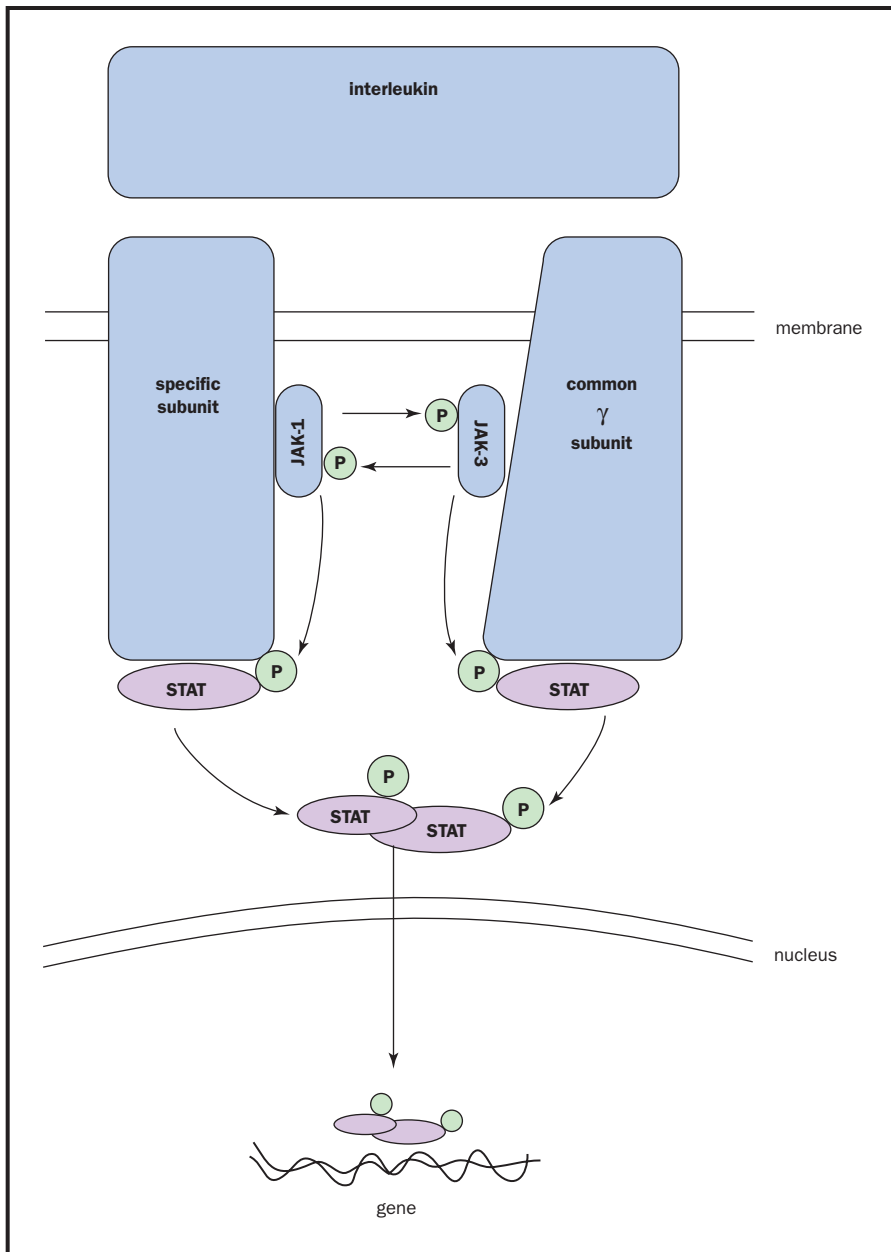
The genes involved in two additional rare forms of SCID, reticular dysgenesis and cartilage-hair hypoplasia, remain unknown. In addition, the gene(s) for over 30 percent of those patients diagnosed with SCID also remain unidentified, even though these patients seem to display the same **phenotypes** as those patients whose genetic defect has been identified.

Gene Therapy

Understanding the genes responsible for SCID has also led to an increased understanding of the genes involved in the overall development of the immune system. It has also helped to unravel the complicated signaling pathways between the cells of the immune system that control and define the immune response itself. There has also been another major benefit. The study of SCID has, in the past, aided in developing an effective program for bone marrow transplantation.

autoimmunity immune reaction to the body's own tissues

phenotypes observable characteristics of an organism



Interleukin binds to its two-part receptor, triggering JAK-1 and JAK-3 to add phosphates to each other and to STAT (signal transducers and activators of transcription). This allows the STATs to combine and enter the nucleus, where they promote gene expression leading to immune system development. SCID can arise from mutations that inactivate either the γ subunit or JAK-3 (shaded structures).

In 1990 the first gene therapy was attempted in two ADA-SCID patients. ADA-SCID was specifically chosen for the first attempted gene therapy for several reasons: bone marrow transplantation had indicated that replacing the defective gene was possible; mouse SCID models had been effectively treated through gene therapy; and ADA-SCID could be treated with some success with enzyme-replacement therapy, enhancing the opportunity for successful gene therapy.

Unfortunately, the majority of ADA-SCID patients, as well as the 3,000 patients enrolled in gene therapy trials for a broad array of other diseases since the first ADA-SCID gene therapy, have not been significantly helped. One exception is the very first gene therapy patient, four-year-old Ashanti de Silva who, ten years later, had a normal lifestyle, with a level of 20 to 25



percent normal T-lymphocytes. Learning from these gene therapy experiences, French researchers modified the procedure. In 2000 they reported successful gene therapy for two infants with SCID-X1. The patients had left the hospital and its protective isolation after a three-month stay. Ten months after gene therapy, they remained healthy, with normal levels of B- and T-lymphocytes and natural killer cells. Thus SCID continues to define the current state of the art for gene therapy, exposing its limitations while simultaneously pointing to its eventual success. SEE ALSO EMBRYONIC STEM CELLS; GENE THERAPY; HIV; IMMUNE SYSTEM GENETICS; RODENT MODELS; SIGNAL TRANSDUCTION.

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Bibliography

Fischer, Alain. "Primary Immunodeficiency Diseases: An Experimental Model for Molecular Medicine." *Lancet* 357, no. 9271 (2001): 1863–1869.

Fischer, Alain, et al. "Gene Therapy for Human Severe Combined Immunodeficiencies." *Immunity* 15 (2001): 1–4.

Leonard, Warren J. "X-linked Severe Combined Immunodeficiency: From Molecular Cause to Gene Therapy within Seven Years." *Molecular Medicine Today* 6, no. 10 (2000): 403–407.

Wheelwright, J. "Body, Cure Thyself." *Discover* 23, no. 2 (2002): 62–68.

Sex Determination

Sex determination refers to the mechanisms employed by organisms to produce offspring that are of two different sexes. First we present an overview of the sex determination mechanisms used by mammals. Then we discuss the great variety of mechanisms used by animals other than mammals.

Mammalian Mechanisms

A developing mammalian embryo's gender is determined by two sequential processes known as primary and secondary sex determination.

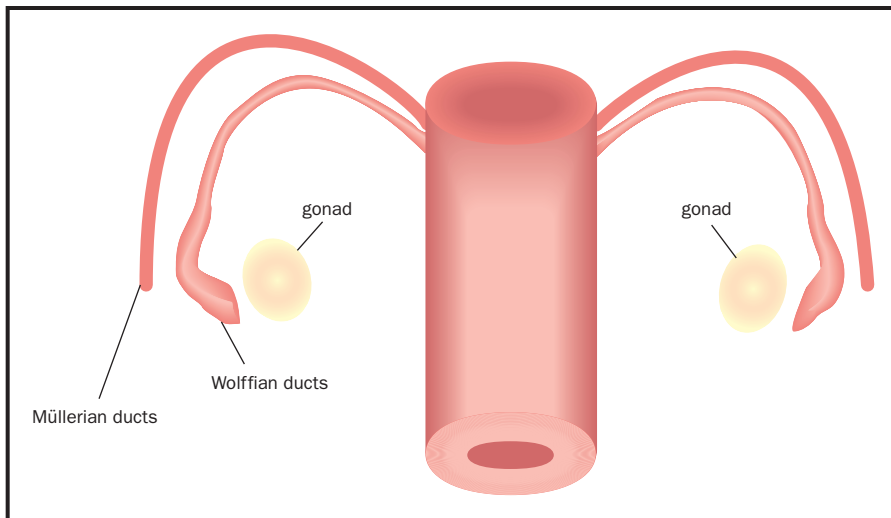
Primary Sex Determination. Early in an embryo's development (four weeks after fertilization, in humans), two groups of cells become organized into the gonad rudiment that will eventually develop into either the ovaries or testicles. These **gonads** will eventually be the source of **gametes** in the adult. However, at this early stage they are unstructured organs that lack sex-specific features but have the potential to develop into gonads.

The first visible indication of sex-specific development, occurring in week seven in humans, is in males, with the gonads restructuring into two distinct compartments: the testicular cords and the interstitial region. In females, the gonads appear to lack distinct structures until later in development. Primary sex determination, including the differentiation of an embryo's gonads, is dependent on genetic factors associated with the embryo's sex chromosomes.

Secondary Sex Determination. Secondary sex determination involves the development of additional sex-specific characteristics, such as the genitalia. This secondary pathway is controlled by sex-specific hormones that are

gonads testes or ovaries

gametes reproductive cells, such as sperm or egg



Early in development, every fetus has two sets of primitive ducts. Under the influence of testosterone and the hormone AMH, the Wolffian ducts develop into the male vas deferens and accessory structures. Without testosterone or AMH, the Wolffian ducts degenerate and the Müllerian ducts develop into the female fallopian tubes, uterus, cervix, and upper vagina.

secreted by the differentiated gonad. These hormones influence the sex differentiation of other parts of the body, including two pairs of ducts present in all developing embryos: the Müllerian ducts and the Wolffian ducts.

Testicles produce Müllerian inhibiting substance, a hormone that causes the Müllerian duct to degenerate. They also produce testosterone, which causes the Wolffian duct to develop into the internal male genitalia, such as the seminal vesicles and the vas deferens. Testosterone also promotes the development of the external male genitalia, including the penis, and it reduces the development of the breasts.

In females, where there are no testicles and where there is therefore no Müllerian inhibiting substance, the Müllerian duct differentiates into internal female genitalia: the **fallopian tubes**, uterus, and cervix.

Discovering the Testis-Determining Factor. The different effects of the primary and secondary sex determination pathways was demonstrated by embryological transplant experiments carried out by Alfred Jost in the 1940s at the Collège de France in Paris, France. When Jost placed an undifferentiated gonad from a male rabbit next to an undifferentiated gonad inside a female fetus, the gonad from the female developed into an ovary, and the gonad from the male developed into a testicle, as it would have done inside the male. Hence, the sex of the gonad was dependent upon its genotype (XX or XY) and is a result of primary sex determination. The genitalia of these experimental animals revealed the influence of secondary sex determination mechanisms. Under the conflicting signal of the two gonads, the Müllerian duct, which normally would have developed into female genitalia, degenerated partially, and the Wolffian duct, which normally would have degenerated, began to develop into male genitalia.

Jost's experiment indicated that the sex differentiation of a gonad is determined by its sex chromosomes, and that the sexual characteristics of other tissue are determined by the gonads, not by the chromosomes in the tissues themselves. Jost also showed that in the absence of either gonad, the fetus develops as a female. Female development, then, is apparently a "default" pathway that can be overridden to produce a male.

fallopian tubes tubes through which eggs pass to the uterus

meiosis cell division that forms eggs or sperm

phenotypic related to the observable characteristics of an organism

transcription messenger RNA formation from a DNA sequence

Studies by C. Ford, P. Jacobs, and co-workers in 1959 demonstrated the importance of genes on the X and Y chromosomes as sexual determinants by documenting the sexual phenotypes of humans with abnormal chromosomal constitutions. Errors in **meiosis** can produce sperm or egg cells that have abnormal numbers of sex chromosomes. Upon fertilization, these cells develop into embryos with their own aberrant sex chromosome dosage.

Cells in individuals with Turner's syndrome contain a single copy of the X chromosome and no copies of the Y chromosome. These "XO" individuals develop as females, although their ovaries are nonfunctional. This demonstrates that just a single copy of the X chromosome is sufficient to direct most female sex development.

The reciprocal condition, "YO," with no X chromosome present, has not been documented in humans, as X chromosomes contain genes necessary for an embryo's survival. Individuals with Klinefelter's syndrome (XXY), however, develop as **phenotypic** males, although they produce no sperm. This indicates that a single copy of the Y chromosome is sufficient to override the female developmental program and promote most male development.

Such observations led to speculation that the Y chromosome contains a "testis-determining factor" necessary to activate development of the male gonads. Several genes on the Y chromosome were suggested as possible testis-determining factors but ultimately rejected. In 1990 Peter Goodfellow and coworkers at the Human Genetics Laboratory in London, England, studied a group of sex-reversed XX males. Such individuals develop as phenotypic males despite being XX individuals.

The researchers discovered that the XX males had a small segment from a Y chromosome incorporated into one of their X chromosomes. This same segment was found to be missing from the Y chromosome of a group of sex-reversed XY individuals, who developed as phenotypic females. The segment acted as a testis-determining factor, as its presence was correlated with the activation of male development. DNA sequencing of the segment identified a gene that was named "**SRY**" from the description "sex-determining region of the Y chromosome."

SRY's Function. Studies in mice have supported *SRY*'s role as a primary determinant of male development. The mouse homolog of *SRY* (*Sry*) is expressed in developing gonads in males but not females. It is present during but not after testis differentiation. Finally, experiments have been conducted where introducing the *SRY* gene into the genomic DNA of embryonic female (XX) mice causes some of them to develop as males.

Despite the clear causal relationship between this gene and male development, the specific mechanisms involved are unclear. The *SRY* protein is similar to "high-mobility group" proteins, which regulate the **transcription** of other genes. Its structure contains a domain that can bind to specific target DNA sequences. Mutations to this domain, which reduce *SRY*'s ability to bind correctly to DNA, are frequently observed in XY individuals that develop as females.

SRY could conceivably function by activating genes involved in testicle differentiation, by repressing genes involved in ovary development, or by

doing both of these things. To discriminate among the possibilities, it is necessary to know more about the next level of genes in the developmental cascade, *SRY*'s targets. Several possible direct targets of *SRY* have been proposed. Most notable is the *SOX9* gene, which also encodes a high-mobility group protein that is known to promote male differentiation and which is expressed in the developing male gonad immediately after *SRY* is first expressed.

Female Development as the Default. When Jost removed the gonads from embryonic rabbits, the embryos that were XX as well as those that were XY developed as females, though they lacked internal genitalia. This finding emphasized that gonads are critical to secondary sex determination and in the absence of male-specific hormones, female characteristics develop even in an XY individual.

As noted above, individuals with only a single copy of the X chromosome in each cell can survive, and they develop as females. Their ovaries develop normally at first but degenerate around the time of birth, resulting in a sterile adult. Hence a single X chromosome suffices for sex determination, but two copies are needed for ovary maintenance.

One explanation is that the female developmental program is the default, with embryos developing as females unless there are alternative instructions. *SRY* is a major switch gene required for male development, and only a gonad whose cells contain a copy of *SRY* will differentiate into a testicle.

Despite intensive searching, no major switch gene has been identified for the female developmental pathway. One candidate, *DAX1*, was proposed as the main ovarian differentiation gene principally because it was found to be duplicated on the X chromosomes of two XY siblings who developed as females. However, experimentally disrupting the mouse homolog of *DAX1* had no effect on the sex determination, maturation, or fertility of female XX mice.

Instead of having a positive regulatory role, *DAX1* appears to be antagonistic to *SRY*'s function. When *DAX1* is present in two copies, as in the XY sisters, it apparently disrupts *SRY* function sufficiently to prevent initiation of the male developmental program. These observations are consistent with the idea that the female sex determination pathway is the default option.

Nonmammalian Mechanisms

Sex determination and differentiation occur in virtually all complex organisms, but the mechanisms used by various animal classes, and even by various vertebrates, differ significantly. Birds, for instance, lack a clear homolog of *SRY*. In birds it is the female, rather than the male, that has two different sex chromosomes, with males being ZZ, and females ZW. In many reptiles, environmental conditions, rather than genetic factors, are the primary determinant of sex. The temperature at which eggs are incubated determines sex in some lizard, turtle, and alligator species.

In both the fruit fly *Drosophila melanogaster* and the nematode *Caenorhabditis elegans*, the primary sex determination mechanisms and the molecular cascades controlling sexual differentiation have been studied

nematode worm of the Nematoda phylum, many of which are parasitic



autosomal describes a chromosome other than the X and Y sex-determining chromosomes

hormones molecules released by one cell to influence another

extensively. Primary sex determination in these animals does not involve the Y chromosome but instead is determined by the ratio of the number of X chromosomes to **autosomal** (nonsex) chromosomes. By examining individuals with unusual numbers of various chromosomes it has been determined that in *Drosophila* those with one or fewer X chromosomes per diploid autosome set develop as males while those with two or more X chromosomes develop as females. Individuals with intermediate ratios such as those with two X chromosomes and a triploid set of autosomes develop as intersexes with both male and female characteristics. Although this ratio serves as the primary determinant of sex in both of these organisms, the specific gene products that influence this ratio assessment are different, demonstrating that different molecular mechanisms can be used for a similar purpose.

There is also significant variability in the strategies by which the outcome of sex determination is communicated to the various tissues that undergo sexual differentiation. In humans and most other mammals, the presence or absence of SRY protein in cells of the gonad specifies their sexual differentiation and which hormones are secreted by the gonad to direct the sexual differentiation of most other cells in the individual.

In *Drosophila* **hormones** have little effect on sexual differentiation. Instead, with only a few exceptions, each cell decides its sex independently of other cells and tissues. This “cell autonomous” mechanism is demonstrated in experimentally produced mosaic organisms called gynandromorphs (“male-female forms”) in which some cells are XX (female) and others XO (male). Such individuals develop into adults with a mix of male and female cell types that match each cell’s genotype. The lack of evolutionary conservation of sex-determining mechanisms among animals is particularly interesting because of the similarities that exist in other major switch genes for basic developmental processes. SEE ALSO ANDROGEN INSENSITIVITY SYNDROME; GENE EXPRESSION: OVERVIEW OF CONTROL; NONDISJUNCTION; TRANSCRIPTION FACTORS; X CHROMOSOME; Y CHROMOSOME.

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Bibliography

- Berta, Philippe, et al. “Genetic Evidence Equating *SRY* and the Testis-Determining Factor.” *Nature* 348 (1990): 448–450.
- Cline, Thomas W., and Barbara J. Meyer. “Vive la Difference: Males vs. Females in Flies vs. Worms.” *Annual Reviews of Genetics* 30 (1996): 637–702.
- Gilbert, Scott F. *Developmental Biology*, 6th ed. Sunderland, MA: Sinauer Associates, 2000.
- Hodgkin, Jonathan. “Genetic Sex Determination Mechanisms and Evolution.” *BioEssays* 4 (1992): 253–261.
- Sinclair, Andrew H., et al. “A Gene from the Human Sex-Determining Region Encodes a Protein with Homology to a Conserved DNA-Binding Motif.” *Nature* 346 (1990): 240–244.
- Zarkower, David. “Establishing Sexual Dimorphism: Conservation amidst Diversity?” *Nature Reviews Genetics* 3 (2001): 175–185.

Sex-Linked Inheritance See *Inheritance Patterns*

Sexual Orientation

The biological basis of sexual orientation (heterosexuality, homosexuality, or bisexuality) has long been a topic of controversy in both science and society. A growing body of research supports the view that genetics and the environment work together to determine sexual orientation. Some issues remain unclear. First, how much of sexual orientation is genetic and how much is shaped by environmental influences, including family, society, and culture? Second, is sexual orientation a fixed trait, or is it subject to environmental influence and changeable over time? Two types of genetic studies, classical family/twin/adoption studies and biological/molecular studies, support multiple genetic and environmental determinants in male and female sexual orientation.

Twin and Family Studies

Twin studies are a classic tool for examining the role of genes. Twins brought up together share a similar environment. **Monozygotic** twins share all their genes, while **dizygotic** twins share only half their genes. Early twin studies by Franz Kallmann in 1952 and Leonard Heston in 1968 reported that if one monozygotic twin was homosexual, there was a greater chance the other twin would be homosexual. The likelihood of this was greater than for dizygotic twins. These studies were potentially biased. They recruited homosexual subjects and had relatively small sample sizes. Recent twin studies have examined all twins in a community without regard to sexual orientation, providing large, less biased sample sizes. In 2000 Kenneth Kendler and colleagues evaluated genetic and environmental factors in a large U.S. sample of twin and nontwin sibling pairs. Sexual orientation was classified as heterosexual or nonheterosexual (bisexual or homosexual) and was determined by a single item on a self-report questionnaire. There was a greater chance for both monozygotic twins to be nonheterosexual than for dizygotic twins or sibling pairs. Results suggested that sexual orientation was greatly influenced by genetic factors, but family environment might also play a role. One problem with this study is that a single item was used to assess the complexity of sexual orientation.

Katherine Kirk's study in 2000 involved a community sample of almost 5,000 adult Australian twins who answered an anonymous questionnaire on sexual behavior and attitudes. Multiple measures of sexual orientation (behaviors, attitudes, feelings) provided stronger evidence for additive genetic influences on sexual orientation. **Heritability estimates** of homosexuality in this sample were 50 to 60 percent in females and 30 percent in males. In 1999 J. Michael Bailey found that if a man was homosexual, the percentage of his siblings who were homosexual or bisexual was 7 to 10 percent for brothers and 3 to 4 percent for sisters, higher than would be due to chance.

Some family studies have reported more homosexuals had homosexual maternal relatives but not paternal relatives. This might support a genetic factor on the X chromosome and/or environmental influences. Other, similar studies did not find this. Thus, evidence exists for both genetic and environmental determinants of sexual orientation which may be different for men and women.

monozygotic genetically identical

dizygotic fraternal or nonidentical

heritability estimates how much of what is observed is due to genetic factors

hypothalamus brain region that coordinates hormone and nervous systems

alleles particular forms of genes

A 2000 study examined whether sexual orientation is fixed or changes with time through environmental influence or the effects of aging. J. Michael Bailey recruited a community sample of twins from the Australian Twin Registry and assessed sexual orientation, childhood gender nonconformity (atypical gender behavior), and continuous gender identity (an individual's self-identification as "male" or "female"). Familial factors were important for all traits, but less successful in distinguishing genetic from shared environmental influences. Only childhood gender nonconformity was significantly heritable for both men and women. Statistical tests suggested that causal factors differed between men and women, and for women provided significant evidence for the importance of genetics factors.

Birth-order studies found homosexual males were not usually first born, having older siblings. Extremely feminine homosexual men had a higher than expected proportion of brothers, not an equal numbers of brothers and sisters.

Biological and Genetic Linkage Studies

Biological studies looking at the **hypothalamus** have found differences between homosexual and heterosexual men and women. Some researchers found differences in parts of the hypothalamus, while others did not. What these findings mean is not clear because they were inconsistent.

In 1995 William Turner examined the ratio of males to females among relatives of the mothers of male homosexuals. He reported that the sex ratio was not normal in maternal relatives. The normal ratio, for relatives of heterosexual males, was an even split: 50 percent male relatives, 50 percent female relatives. The number of male relatives of homosexual males, on the other hand, was significantly lower than the number of female relatives. Also, 65 percent of the mothers of homosexuals had no live-born brothers, or else they had only one live-born brother. On the paternal side, however, the number of male and female relatives of male homosexuals was same as that found for heterosexuals, and the sex ratio of relatives on both the maternal and paternal side for female homosexuals was the same as for heterosexuals. These findings would support genetic factors on the X chromosome, which males inherit from their mothers, as a factor that may cause fetal or neonatal loss of males.

Molecular studies found a linkage between male homosexuality and the X chromosome. Dean Hamer and colleagues in 1993 and Nan Hu and colleagues in 1995 conducted DNA linkage analyses in U.S. families with two homosexual brothers. There was significant linkage at Xq28 for 64 percent of homosexual male siblings but not for homosexual females (Xq28 is band 28 of the long arm of the X chromosome). George Rice and colleagues in 1999 examined four **alleles** at Xq28 in fifty-two Canadian male homosexual siblings but did not find any such linkage. This could represent genetic variation, diagnostic differences, and/or different methods of data analysis.

In summary, family, biological, and molecular data support multiple genetic and environmental bases for sexual orientation, and evidence exists for childhood gender nonconformity. SEE ALSO COMPLEX TRAITS; GENE AND THE ENVIRONMENT; PUBLIC HEALTH, GENETIC TECHNIQUES IN; NOMENCLATURE; SEX DETERMINATION; STATISTICS; TWINS; X CHROMOSOME; Y CHROMOSOME.

Harry Wright and Ruth Abramson

Bibliography

- Bailey, J. Michael., M. P. Dunne, and N. G. Martin. "Genetic and Environmental Influences on Sexual Orientation and Its Correlates in an Australian Twin Sample." *Journal of Personal and Social Psychology* 78 (2000): 524–536.
- Friedman, R. C., and J. I. Downey. "Homosexuality." *New England Journal of Medicine* 331 (1994): 923–930.
- Hamer, Dean H., et al. "A Linkage between DNA Markers on the X Chromosome and Male Sexual Orientation." *Science* 261 (1993): 321–327.
- Kendler, Kenneth S., et al. "Sexual Orientation in a U.S. National Sample of Twin and Nontwin Sibling Pairs." *American Journal of Psychiatry* 157 (2000): 1843–1846.

Sickle Cell Disease *See Hemoglobinopathies*

Signal Transduction

To survive, an organism must constantly adjust its internal state to changes in the environment. To track environmental changes, the organism must receive signals. These may be in the form of chemicals, such as **hormones** or nutrients, or may take another form, such as light, heat, or sound. A signal itself rarely causes a simple, direct chemical change inside the cell. Instead, the signal sets off a chain of events that may involve several or even dozens of steps. The signal is thereby transduced, or changed in form. Signal transduction refers to the entire set of pathways and interactions by which environmental signals are received and responded to by single cells.

Signal transduction systems are especially important in multicellular organisms, because of the need to coordinate the activities of hundreds to trillions of cells. Multicellular organisms have developed a variety of mechanisms allowing very efficient and controlled cell-to-cell communication. Though we take it for granted, it is actually astonishing that our skin, for example, continues to grow at the right rate to replace the continuous loss of its surface every day of our lives. This tight regulation is found in every tissue of our body all of the time, and when this fine control breaks down, cancer may be the result. Clearly the molecular mechanisms behind this astounding level of control must be powerful, versatile, and sophisticated.

Signals, Receptors, and Cascades

The signals that cells use to communicate with one another are often small amino acid chains, called **peptides**. Depending on the cell type that releases them and the effect they have on the target cell, they may be called hormones, growth factors, neuropeptides, **neurotransmitters**, or **cytokines**. Other small molecules can also be signals, such as amino acids and steroids such as testosterone. External signals such as odorants and tastes can be carried to us in the atmosphere or in the fluids of our food and drinks. Stretch, pressure, and other mechanical effects as well as heat, pain, and light can also act as signals.

Given the huge variety of signals to which a cell is exposed, how does it know which to respond to? The answer is that signals are received by protein receptors made by the cell, and a cell is sensitive only to those signals for which it has made receptors. For instance, every cell in the body is

hormones molecules released by one cell to influence another

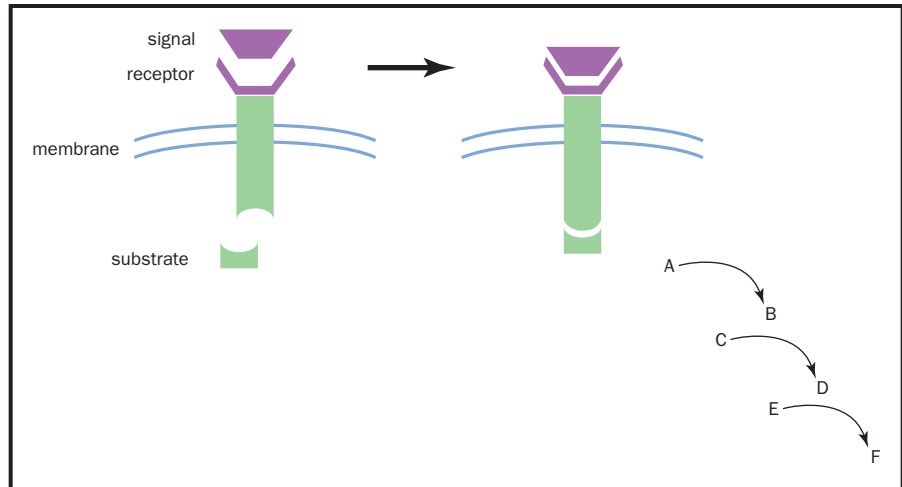
peptides amino acid chain

neurotransmitters molecules released by one neuron to stimulate or inhibit a neuron or other cell

cytokines immune system signaling molecules



Signal transduction converts an environmental signal into a cellular response. A cascade allows rapid response, amplification and control.



exposed to estrogen circulating in the blood, but only a subset of them make estrogen receptors, and are therefore sensitive to its influence.

Chemical signals such as hormones bind to their receptors, usually at the surface of the cell (the plasma membrane), but sometimes within the cell. This causes a conformation (shape) change in the receptor. The conformation change typically alters the ability of the receptor to bind to another molecule in the cell, modifying that molecule's conformation, or triggering other actions.

This sequence of events triggered by the signal-receptor interaction is called a **transduction** cascade. A transduction cascade involves a network of **enzymes** that act on one another in specific ways to ultimately generate precise and appropriate responses. These responses may include alterations in cell motility or division, induction of the expression of specific genes, and the regulation of **apoptosis**. The molecular details of several such cascades are known, but many more undoubtedly remain to be discovered.

The value of this complex cascade of events is severalfold. First, the network of interactions provides many levels of control, so that the cell can tailor the magnitude and timing of its response very finely. Second, the many levels of interaction allow amplification of the original signal to quickly produce a strong response to a small stimulus. For example, there may be only a few hundred copies of a specific receptor on the surface of a typical cell. Activation of even a small percentage of them, acting through these amplifying enzyme cascades, can result in activation of millions of downstream target molecules. This explains how even very small amounts of signals such as growth factor can have such profound effects on appropriately receptive cells.

The Importance of Phosphorylation and Dephosphorylation

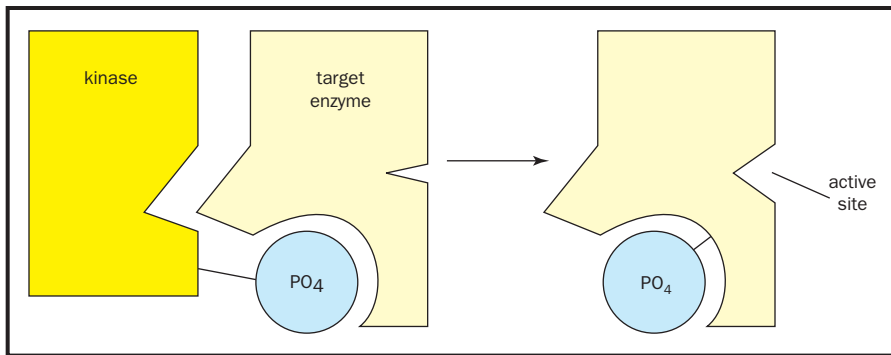
After a signal is received, signal transduction involves altering the behavior of proteins in the cascade, in effect turning them on or off like a switch. Adding or removing phosphates is a fundamental mechanism for altering the shape, and therefore the behavior, of a protein. **Phosphorylation** may open up an enzyme's active site, allowing it to perform chemical reactions,

transduction conversion of a signal of one type into another type

enzymes proteins that control a reaction in a cell

apoptosis programmed cell death

phosphorylation addition of the phosphate group PO_4^{3-}



Phosphorylation may open the active site of an enzyme.

or it may frequently generate a binding site allowing a specific interaction (may make a bulge in one side preventing the protein from fitting together) with a molecular partner.

Enzymes that add phosphate groups to other molecules are called **kinases**, and the molecules the enzymes act on are called substrates. Protein kinases are a family of enzymes that use **ATP** to add phosphate groups on to other proteins, thereby altering the properties of these substrate proteins. Protein kinases themselves are frequently turned on or off by phosphorylation performed by other protein kinases; thus a kinase can be both enzyme and substrate.

Protein kinases can be broadly divided into two groups based on the amino acids to which they add phosphate groups. The serine/threonine kinases (ser/thr) are found in all **eukaryotic** cells and tend to be involved in regulation of metabolic and cytoskeletal activity. The tyrosine (tyr) kinases are found in all animals but not in yeast, protozoa, or plants, and appear to have evolved specifically to deal with the complex challenges of signaling in animals.

Molecular switches are useful only if they can also be flipped back to their original state. This is achieved by specific protein phosphatases, which can remove phosphate groups from kinase substrates.

Signal Transduction: The RTK Pathway

How does the cell “know” when a particular receptor molecule in the membrane is occupied, and how is that information chemically translated into actions within the cell? Let us examine the signaling pathway for the receptor tyrosine kinases (RTK). The RTKs are a very powerful and important family of signaling molecules and include receptors for potent growth factors and such hormones as insulin, epidermal growth factor, and nerve growth factor.

In this system, the extracellular “**ligand**” (growth factor or hormone) must crosslink two receptor molecules in order to begin the transduction cascade. The interaction of the two intracellular domains of the receptors then initiates a signaling response.

The simplest RTKs have three parts: a ligand binding site outside the cell, a single membrane-spanning domain, and a tyrosine kinase domain inside the cell. The ligand is typically a diffusible peptide or small protein

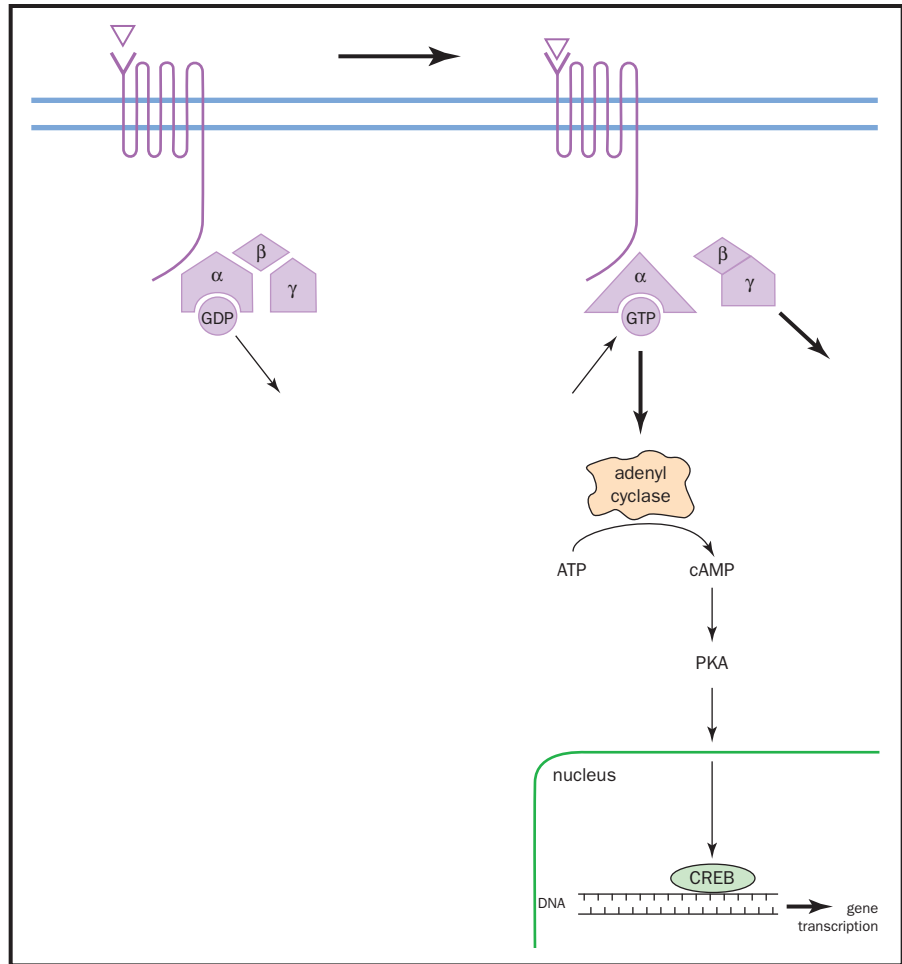
kinases enzymes that add a phosphate group to another molecule, usually a protein

ATP adenosine triphosphate, a high-energy compound used to power cell processes

eukaryotic describing an organism that has cells containing nuclei

ligand a molecule that binds to a receptor or other molecule

In the GPCR pathway, an external signal causes an internal, multi-part G protein to exchange GDP for GTP. It then activates adenylyl cyclase and ultimately triggers gene transcription.



produced elsewhere in the organism, and in this case is the specific growth factor recognized by the RTK. In the absence of its specific growth factor, this receptor remains unbound to a second receptor, and is inactive. Growth factor, when it arrives, binds to two receptors, cross-linking them. This causes the two tyrosine kinase domains to come into contact with one another. Each kinase now has a substrate, formed by the other receptor, and so each phosphorylates the other on multiple tyrosine sites.

The receptors may now bind with one or more other proteins (called SH2 proteins) that specifically recognize their phosphorylated tyrosines. Many of these are enzymes, while others are adapter molecules that in turn attract and bind other enzymes. Often these enzymes are inactive until they join the receptor complex, because their substrates are found only in the membrane. The products of these enzymes may act on yet other molecules, thus continuing the signaling cascade, or they may be used in cell metabolism for growth or other responses.

Signal Transduction: The GPCR Pathway and G Proteins

The G protein coupled receptors (GPCR) illustrate another way to receive and interpret a signal, in which the ligand binds to a single transmembrane protein, causing a conformation change and activating “G proteins” inside

the cell. G proteins are so named because they have a binding site for guanine nucleotide, either a diphosphate (GDP) or a triphosphate (GTP). Like ATP, GTP is the high-energy form of the nucleotide, while GDP is the low-energy form.

The GPCR family of proteins has at least 300 members in humans. They are specific receptors for numerous neurotransmitters, hormones, peptides and other substances. A large subgroup of these are odorant receptors directly responsible for our senses of taste and smell.

Surprisingly, a further group of GPCRs is responsible for our sense of vision. Opsin molecules, including rhodopsin, are actually GPCRs. Instead of a ligand binding site, rhodopsin has a molecule of retinal bound in the same relative position. Light alters the conformation of the retinal, and rhodopsin responds to this by altering its protein conformation. This causes G protein activation in a similar manner to the other GPCRs.

Each GPCR is associated with a particular type of G protein inside the cell, and there are dozens of different G proteins known. G proteins are generally inactive in the GDP-bound form. They are activated when GDP departs and is replaced by GTP. GTP is found in excess in cells, and so GDP departure is followed rapidly by GTP binding. This causes a profound conformational change, allowing the G protein to interact with and influence numerous target molecules.

Each GPCR associated G protein consists of three parts, the α , β , and γ subunits. The α subunit binds either GDP or GTP. The β and γ subunits are always found together in a complex called $G\beta\gamma$. In unstimulated cells the whole three-part complex is found in the **plasma membrane**, with GDP in the binding site.

Binding of ligand to the GPCR causes a conformational change that is transmitted to the cytoplasmic region of the receptor, which interacts with the G protein to dissociate the GDP. This results in GTP binding, which alters the structure of the α subunit, freeing it from $G\beta\gamma$. Both parts, the $G\alpha(\text{GTP})$ and the $G\beta\gamma$, now diffuse away from the receptor and separately interact with and influence many other molecules in the cell.

Second Messengers

One target for certain $G\alpha(\text{GTP})$ types is the enzyme adenylyl cyclase. This enzyme uses ATP to generate cyclic AMP (cAMP). The cAMP molecule is a “second messenger,” one of a family of small diffusible substances that powerfully induce cytoplasmic responses.

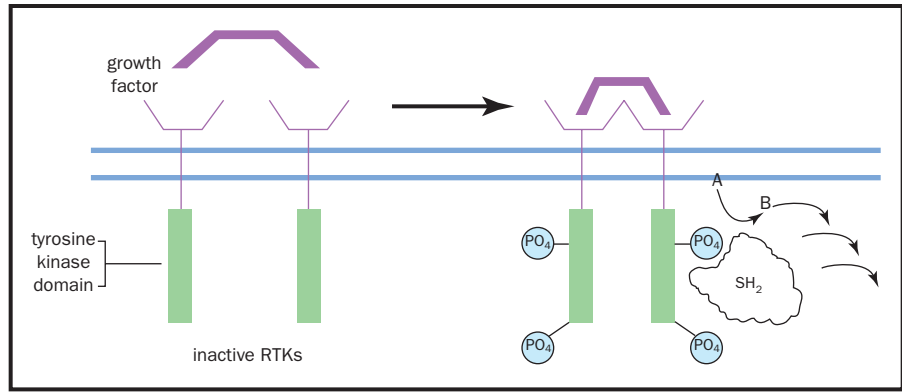
Cyclic AMP exerts much of its effects by activating the cAMP-dependent protein kinase (PKA), a ser/thr kinase that can phosphorylate and influence many cellular proteins. For example, PKA phosphorylates CREB (cyclic AMP response element binding protein), which is found attached to the promoters of many genes. Phosphorylation of CREB by PKA can thus regulate the expression of these genes.

Another $G\alpha(\text{GTP})$ target is an enzyme, phospholipase C, which cleaves a membrane lipid called PIP-2. This produces two products: DAG and IP-3. DAG stays in the membrane and binds all members of the ser/thr protein kinase C (PKC) family of enzymes, which may then become activated

plasma membrane
outer membrane of the
cell



Activated RTK binds SH₂ proteins, which can then act as their substrates (A) to form products (B).



endoplasmic reticulum
network of membranes
within the cell

and phosphorylate and regulate a host of metabolic and structural enzymes. IP-3, another second messenger, rapidly diffuses to IP-3 receptors in the **endoplasmic reticulum** membrane. When IP-3 binds, it opens channels in the membrane, releasing stored calcium into the cytoplasm. Calcium is normally kept at very low levels in the cytoplasm, and even small increases cause numerous major effects, so that calcium is also regarded as a powerful second messenger. These effects include the activation of various calcium binding proteins such as calmodulin and its many relatives. The calcium-bound versions of these proteins regulate many other enzymes. For example, calcium activates all members of a large and important family of ser/thr kinases called calcium/calmodulin dependent protein kinases (CAM kinases), which themselves regulate the activity of numerous important substrate molecules.

Just as protein kinases need to be turned off, so too do G proteins. This occurs when the GTP on the G α is cleaved to generate GDP, thus favoring the reformation of the three-part inactive protein.

Interacting Pathways, Defective Signaling, and Treatments for Disease

The GPCR and RTK pathways do not necessarily remain separate, either from each other or from other signaling pathways. One of the most important pathways is directly downstream of RTKs and also many GPCRs, and is called the mitogen activated protein (MAP) kinase cascade. MAP kinase is an abundant ser/thr kinase that, when activated, phosphorylates and powerfully affects the activity of a large number of cytoskeletal, signaling, and nuclear proteins, including an important family of **transcription factors**, thus directly influencing gene expression.

Both pathways also help regulate a particularly important process called **apoptosis**. Many cells need specific growth factors to stay alive; the growth factors trigger pathways involving various ser/thr protein kinases that ultimately inactivate molecules that would otherwise promote apoptosis.

The entire signal transduction system normally works astonishingly well, but serious problems can occur. Cancer is unregulated cell growth and occurs when the machinery tightly regulating cell growth breaks down. It is often easy to see how this has occurred. Mutations in growth factor receptors, G proteins, MAP kinases, and other molecules frequently contribute to cancer, and generally result in these molecules losing their normal switching

transcription factors
proteins that increase
the rate of gene trans-
cription

apoptosis programmed
cell death

function, staying in the activated form and therefore inappropriately stimulating these important enzyme cascades.

The complexity of the signaling system makes for challenging research, but once understood it holds the promise for better treatments for cancer and other diseases. This is because each step in each pathway provides one or more targets for drugs. Designing a drug that could quiet the excess signaling caused by defective MAP kinase, for example, might provide a promising cancer treatment.

The examples given thus far provide only an outline of how signal transduction cascades work and an overview of a few of the most important enzymes. The actual process is much more complex, and there is much about the process that remains mysterious. Perhaps the biggest mystery is how the cell makes sense of all of the input from different growth factors, hormones, extracellular substrates, and so on to produce an appropriate response. The solution to this problem will result from a complete understanding and computer modeling of the biochemical and kinetic properties of the components of all these signaling cascades. SEE ALSO APOPTOSIS; CELL, EUKARYOTIC; CELL CYCLE; GENE; GENE EXPRESSION: OVERVIEW OF CONTROL; PHARMACOGENETICS AND PHARMACOGENOMICS.

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Bibliography

Alberts, Bruce., et al. *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.

Scott, John D., and Tony Pawson. "Cell Communication: The Inside Story." *Scientific American* (June 2000).

Special Issue on Mapping Cellular Signaling. *Science* 296, no. 5573 (May 31, 2002).

Speciation

Speciation is the process by which new species of organisms arise. Earth is inhabited by millions of different organisms, all of which likely arose from one early life-form that came into existence about 3.5 billion years ago. It is the task of **taxonomists** to decide which out of the multitude of different types of organisms should be considered species. The wide range in the characteristics of individuals within groups makes defining a species more difficult. Indeed, the definition of species itself is open to debate.

taxonomists scientists who identify and classify organisms

Concepts of Species

In the broadest sense, a species can be defined as a group of individuals that is "distinct" from another group of individuals. Several different views have been put forward about what constitutes an appropriate level of difference. Principal among these views are the biological-species concept and the morphological-species concept.

The biological-species concept delimits species based on breeding. Members of a single species are those that interbreed to produce fertile offspring or have the potential to do so. The morphological-species concept (from the ancient Greek root "morphos," meaning form) is based on classifying species by a difference in their form or function. According to this

concept, members of the same species share similar characteristics. Species that are designated by this criteria are known as a morphological species.

Organisms within a species do not necessarily look identical. For example, the domestic dog is considered to be one species, even though there is a huge range in size and appearance among the different breeds. For naturally occurring populations of organisms that we are much less familiar with, it is much more difficult to recognize the significance of any character differences observed. Therefore deciding what characteristics should be used as criteria to designate a species can be difficult.

Speciation Mechanisms: Natural Selection and Genetic Drift

Before the development of the modern theory of evolution, a widely held idea regarding the diversity of life was the “typological” or “essentialist” view. This view held that a species at its core had an unchanging perfect “type” and that any variations on this perfect type were imperfections due to environmental conditions. Charles Darwin (1809–1882) and Alfred Russel Wallace (1823–1913) independently developed the theory of evolution by natural selection, now commonly known as Darwinian evolution.

The theory of Darwinian evolution is based on two main ideas. The first is that heritable traits that confer an advantage to the individual that carries them will become more widespread in a population through natural selection because organisms with these favorable traits will produce more offspring. Since different environments favor different traits, Darwin saw that the process of natural selection would, over time, make two originally similar groups become different from one another, ultimately creating two species from one. This led to the second major idea, which is that all species arise from earlier species, therefore sharing a common ancestor.

When so much change occurs between different groups that they are morphologically distinct or no longer able to interbreed, they may be considered different species; this process is known as speciation. A species as a whole can transform over time into a new species (vertical evolution) or split into more separate populations, each of which may develop into new species (adaptive radiation).

Modern population geneticists recognize that natural selection is not the only factor causing genetic change in a population over time. **Genetic drift** is the random change in the genetic composition of a small population over time, due to an unequal genetic contribution by individuals to succeeding generations. It is thought that genetic drift can result in new species, especially in small isolated populations.

Isolating Mechanisms

Whether natural selection and genetic drift lead to new species depends on whether there is restricted gene flow between different groups. Gene flow is the movement of genes between separate populations by migration of individuals. If two populations remain in contact, gene flow will prevent them from becoming separate species (though they may both develop into a new species through vertical evolution).

genetic drift evolutionary mechanism, involving random change in gene frequencies

Gene flow is restricted through geographic effects such as mountain ranges and oceans, leading to geographic isolation. Gene flow can also be prevented by biological factors known as isolating mechanisms. Biological isolating mechanisms include differences in behavior (especially mating behavior), and differences in habitat use, both of which lead to a decrease in mating between individuals from different groups.

When geographic separation plays a role in speciation, this is known as allopatric speciation, from the Greek roots *allo*, meaning separate, and “patric,” meaning country. In allopatric speciation, natural selection and genetic drift can act together.

For example, imagine a mud slide that causes a river to back up into a valley, separating a population of rodents into two, one restricted to the shady side of the river, the other to the sunny side. Because coat thickness is a genetically inherited trait, eventually, through natural selection, the population of animals on the cooler side may develop thicker coats. After many generations of separation, the two groups may look quite different and may have evolved different behaviors as well, to allow them to survive better in their respective habitats. Genetic drift may occur especially if either or both populations remain small. Eventually these two populations may be so different as to warrant designation as different species.

It is also possible for new species to form from a single population without any geographic separation. This is known as “ecological” or “sympatric” (from the Greek root *sym*, meaning same) speciation, and it results in ecological differences between morphologically similar species inhabiting the same area. Sympatric speciation can occur in flowering plants in a single generation, due to the formation of a polyploid. **Polyploidy** is the complete duplication of an organism’s genome, for example from n chromosomes to $4n$. Even higher multiples of n are possible. This increase in a plant’s DNA content makes it reproductively incompatible with other individuals of its former species. SEE ALSO CHROMOSOMAL ABERRATIONS; CONSERVATION BIOLOGY: GENETIC APPROACHES; MUTATION; POPULATION GENETICS; SELECTION.

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Bibliography

Futuyma, Douglas J. *Evolutionary Biology*, 3rd ed. Sunderland, MA: Sinauer Associates, 1998.

Mayr, Ernst. *Evolution and the Diversity of Life: Selected Essays*. Cambridge, MA: Belknap Press, 1976.

Statistical Geneticist

Statistical geneticists are highly trained scientific investigators who are specialists in both statistics and genetics. Training in both statistics and genetics is necessary, as the nature of the work is highly interdisciplinary. Statistical geneticists must be able to understand molecular and clinical genetics, as well as mathematics and statistics, to effectively communicate with scientists from these disciplines.

polyploidy presence of multiple copies of the normal chromosome set

Typical statistical geneticists spend much of their time working with computers, since much of the statistical analysis of data is now conducted using computer software instead of pencil and paper. Many statistical geneticists are actively engaged in developing new statistical methods for problems that are specific to genetics. Computers are invaluable tools, since much of the initial evaluation of new statistical methods is performed using computer simulations. That is, artificial data are generated using computer algorithms, and the statistical method is evaluated for its ability to identify the specific genetic effects that were simulated. Once a new statistical method is validated using computer simulations it can then be applied to the analysis of real genetic data.

Given the highly interdisciplinary field of statistical genetics, training in multiple different scientific disciplines is necessary. Many statistical geneticists receive four years of undergraduate training in mathematics, statistics, physics, computer science, or some other analytical field of study. This analytical preparation is often necessary for graduate studies in statistical genetics. It is certainly possible, and even desirable in many cases, to receive undergraduate training in biological sciences and then go on to study statistics and genetics in graduate school. There are many different educational paths that a statistical geneticist can take, but the key is interdisciplinary training in an analytical field and genetics. Given the increasing demand for statistical geneticists in the job market, there are many graduate programs that now specialize in statistical genetics.

After finishing four to six years of graduate training, one can follow any of several career paths. Many graduates receive additional training through postdoctoral studies. These studies, known as a postdoc, can last from one to four years and involve working with a research team to gain additional experience in a particular area. For example, if a student had focused primarily on training in mathematics and statistics, it would be possible to do a postdoc to receive the necessary additional training in genetics.

Students who want to target a particular disease area such as cancer for their career might seek out a postdoc that provides experience in cancer research. Students who feel adequately prepared for the job market from their graduate training may be able to go right into a faculty position at a university or to a scientist position in industry without postdoctoral training. Statistical geneticists are in high demand, and there are often more job openings than qualified people to fill them. This will certainly be a growth profession for the next decade.

Statistical genetics is a very exciting professional area because it is so new and there is so much demand. It is a rapidly changing field, and there are many fascinating scientific questions that need to be addressed. Additionally, given the interdisciplinary nature of statistical genetics, there are plenty of opportunities to interact with researchers and clinicians in other fields, such as **epidemiology**, biochemistry, physiology, **pathology**, evolutionary biology, and anthropology. The salary range for statistical geneticists is quite good, since there is so much demand. This is especially true for university medical schools, where starting annual salaries ranged from \$70,000 to \$80,000 in 2002. Starting salaries for industry positions such as those in pharmaceutical or biotechnology companies were as much as 25 or 50 percent higher.

epidemiology study of incidence and spread of diseases in a population

pathology disease process

A related discipline is genetic epidemiology. Just as statistical genetics requires a combination of training in statistics and genetics, genetic epidemiology requires training in epidemiology and genetics. Since both disciplines require knowledge of statistical methods, there is significant overlap. A primary difference is that statistical geneticists are often more interested in the development and evaluation of new statistical methods, whereas genetic epidemiologists focus more on the application of statistical methods to biomedical research problems. Statistical genetics is an exciting and rewarding career choice for those who have interest and aptitude in the analytical sciences as well as in genetics and biology. SEE ALSO STATISTICS.

Jason H. Moore

Bibliography

Internet Resource

The Committee of Presidents of Statistical Societies (COPSS) Presents Careers in Statistics. American Statistical Association. <<http://www.amstat.org/CAREERS/COPSS/>>.

Statistics

Statistics is the set of mathematical tools and techniques that are used to analyze data. In genetics, statistical tests are crucial for determining if a particular chromosomal region is likely to contain a disease gene, for instance, or for expressing the certainty with which a treatment can be said to be effective.

Statistics is a relatively new science, with most of the important developments occurring with the last 100 years. Motivation for statistics as a formal scientific discipline came from a need to summarize and draw conclusions from experimental data. For example, Sir Ronald Aylmer Fisher, Karl Pearson, and Sir Francis Galton each made significant contributions to early statistics in response to their need to analyze experimental agricultural and biological data. For example, one of Fisher's interests was whether crop yield could be predicted from meteorological readings. This problem was one of several that motivated Fisher to develop some of the early methods of data analysis. Much of modern statistics can be categorized as exploratory data analysis, point estimation, or hypothesis testing.

The goal of exploratory data analysis is to summarize and visualize data and information in a way that facilitates the identification of trends or interesting patterns that are relevant to the question at hand. A fundamental exploratory data-analysis tool is the histogram, which describes the frequency with which various outcomes occur. Histograms summarize the distribution of the outcomes and facilitate the comparison of outcomes from different experiments. Histograms are usually plotted as bar plots, with the range of outcomes plotted on the x-axis and the frequency of the individual outcome represented by a bar on the y-axis. For instance, one might use a histogram to describe the number of people in a population with each of the different genotypes for the *ApoE* alleles, which influence the risk of Alzheimer's disease.

alleles particular forms of genes

The range of outcomes from an experiment are also described mathematically by their central tendency and their dispersion. Central tendency is a measure of the center of the distribution. This can be characterized by the mean (the arithmetic average) of the outcomes or by the median, which is the value above and below which the number of outcomes is the same. The mean of 3, 4, and 8 is 5, whereas the median is 4. The median length of response to a gene therapy trial might be 30 days, meaning as many people had less than 30 days' benefit as had more than that. The mean might be considerably more—if one person benefited for 180 days, for instance.

Dispersion is a measure of how spread out the outcomes of the random variable are from their mean. It is characterized by the variance or standard deviation. The spread of the data can often be as important as the central tendency in estimating the value of the results. For instance, suppose the median number of errors in a gene-sequencing procedure was 3 per 10,000 bases sequenced. This error rate might be acceptable if the range that was found in 100 trials was between 0 and 5 errors, but it would be unacceptable if the range was between 0 and 150 errors. The occasional large number of errors makes the data from any particular procedure suspect.

Another important concept in statistics is that of populations and samples. The population represents every possible experimental unit that could be measured. For example, every zebra on the continent of Africa might represent a population. If we were interested in the mean genetic diversity of zebras in Africa, it would be nearly impossible to actually analyze the DNA of every single zebra; neither can we sequence the entire DNA of any individual. Therefore we must take a random selection of some smaller number of zebras and some smaller amount of DNA, and then use the mean differences among these zebras to make inferences about the mean diversity in the entire population.

Any summary measure of the data, such as the mean or variance in a subset of the population, is called a sample statistic. The summary measure of the entire group is called a population parameter. Therefore, we use statistics to estimate parameters. Much of statistics is concerned with the accuracy of parameter estimates. This is the statistical science of point estimation.

The final major discipline of statistics is hypothesis testing. All scientific investigations begin with a motivating question. For example, do identical twins have a higher likelihood than fraternal twins of both developing alcoholism?

From the question, two types of hypotheses are derived. The first is called the null hypothesis. This is generally a theory about the value of one or more population parameters and is the status quo, or what is commonly believed or accepted. In the case of the twins, the null hypothesis might be that the rates of concordance (i.e., both twins are or are not alcoholic) are the same for identical and fraternal twins. The alternate hypothesis is generally what you are trying to show. This might be that identical twins have a higher concordance rate for alcoholism, supporting a genetic basis for this disorder. It is important to note that statistics cannot prove one or the other hypothesis. Rather, statistics provides evidence from the data that supports one hypothesis or the other.

Much of hypothesis testing is concerned with making decisions about the null and alternate hypotheses. You collect the data, estimate the parameter, calculate a test statistic that summarizes the value of the parameter estimate, and then decide whether the value of the test statistic would be expected if the null hypothesis were true or the alternate hypothesis were true. In our case, we collect data on alcoholism in a limited number of twins (which we hope accurately represent the entire twin population) and decide whether the results we obtain better match the null hypothesis (no difference in rates) or the alternate hypothesis (higher rate in identical twins).

Of course, there is always a chance that you have made the wrong decision—that you have interpreted your data incorrectly. In statistics, there are two types of errors that can be made. A type I error is when the conclusion was made in favor of the alternate hypothesis, when the null hypothesis was really true. A type II error refers to the converse situation, where the conclusion was made in favor of the null hypothesis when the alternate hypothesis was really true. Thus a type I error is when you see something that is not there, and a type II error is when you do not see something that is really there. In general, type I errors are thought to be worse than type II errors, since you do not want to spend time and resources following up on a finding that is not true.

How can we decide if we have made the right choice about accepting or rejecting our null hypothesis? These statistical decisions are often made by calculating a probability value, or p-value. P-values for many test statistics are easily calculated using a computer, thanks to the theoretical work of mathematical statisticians such as Jerzy Neyman.

A p-value is simply the probability of observing a test statistic as large or larger than the one observed from your data, if the null hypothesis were really true. It is common in many statistical analyses to accept a type I error rate of one in twenty, or 0.05. This means there is less than a one-in-twenty chance of making a type I error.

To see what this means, let us imagine that our data show that identical twins have a 10 percent greater likelihood of being concordant for alcoholism than fraternal twins. Is this a significant enough difference that we should reject the null hypothesis of no difference between twin types? By examining the number of individuals tested and the variance in the data, we can come up with an estimate of the probability that we could obtain this difference by chance alone, even if the null hypothesis were true. If this probability is less than 0.05—if the likelihood of obtaining this difference by chance is less than one in twenty—then we reject the null hypothesis in favor of the alternate hypothesis.

Prior to carrying out a scientific investigation and a statistical analysis of the resulting data, it is possible to get a feel for your chances of seeing something if it is really there to see. This is referred to as the power of a study and is simply one minus the probability of making a type II error. A commonly accepted power for a study is 80 percent or greater. That is, you would like to know that you have at least an 80 percent chance of seeing something if it is really there. Increasing the size of the random sample from the population is perhaps the best way to improve the power of a study. The closer your sample is to the true population size, the more likely you are to see something if it is really there.

Thus, statistics is a relatively new scientific discipline that uses both mathematics and philosophy for exploratory data analysis, point estimation, and hypothesis testing. The ultimate utility of statistics is for making decisions about hypotheses to make inferences about the answers to scientific questions. SEE ALSO GENE DISCOVERY; GENE THERAPY: ETHICAL ISSUES; STATISTICAL GENETICIST; TWINS.

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Bibliography

Gonick, Larry, and Woollcott Smith. *The Cartoon Guide to Statistics*. New York: Harper Collins, 1993.

Jaisingh, Lloyd R. *Statistics for the Utterly Confused*. New York: McGraw-Hill, 2000.

Salsberg, David. *The Lady Tasting Tea: How Statistics Revolutionized Science in the Twentieth Century*. New York: W. H. Freeman, 2001.

Internet Resource

HyperStat Online: An Introductory Statistics Book and Online Tutorial for Help in Statistics Courses. David M. Lane., ed. <<http://davidmlane.com/hyperstat/>>.

Stem Cells See Embryonic Stem Cells



lipid fat or waxlike molecule, insoluble in water

enzyme a protein that controls a reaction in a cell

retina light-sensitive layer at the rear of the eye

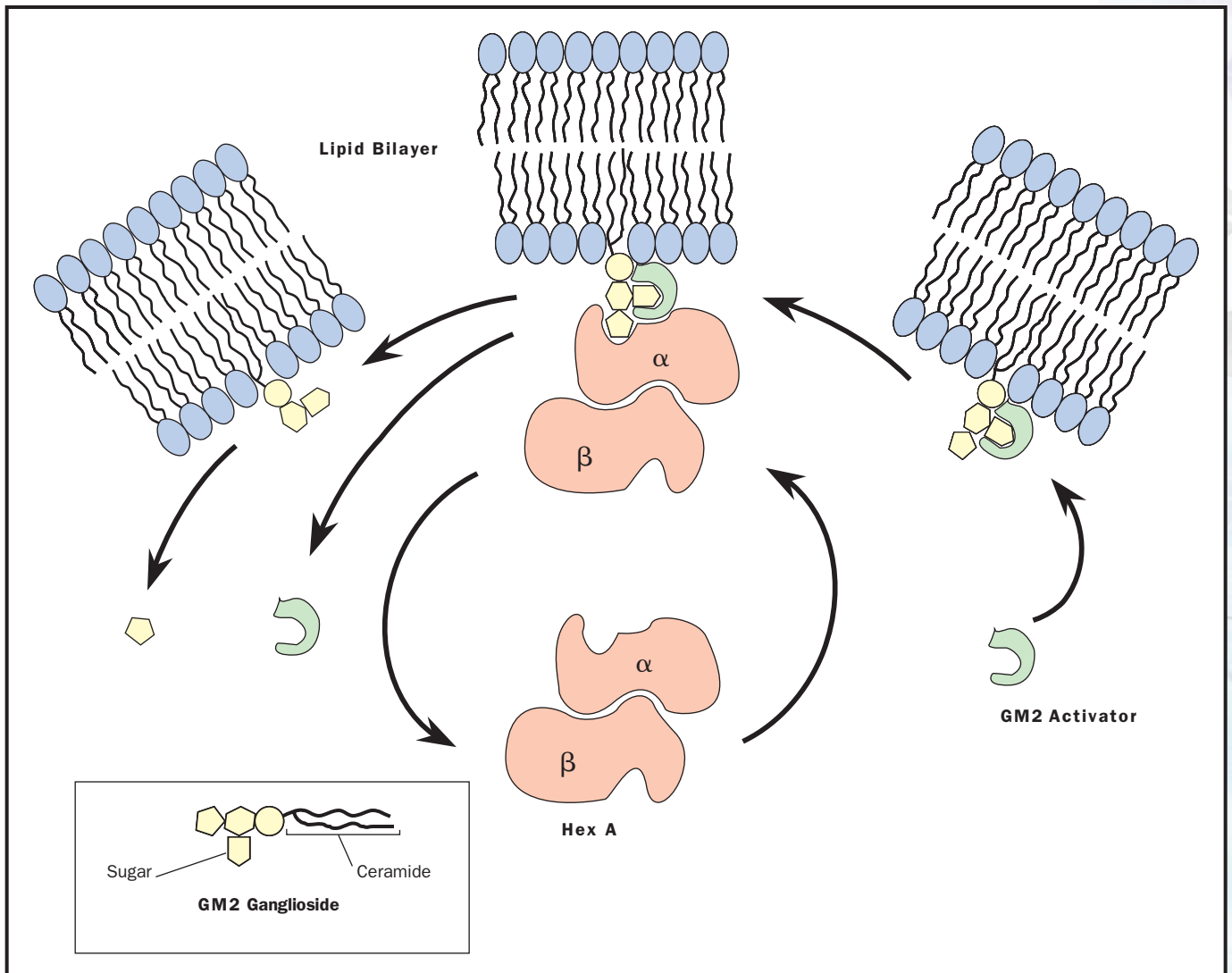
Tay-Sachs Disease

Tay-Sachs disease is a severe genetic disease of the nervous system that is nearly always fatal, usually by three to four years of age. It is caused by mutations in the *HEXA* gene, which codes for a component of the enzyme β -hexosaminidase A or “Hex A.” The resulting accumulation of a brain **lipid** called G_{M2} ganglioside produces brain and spinal cord degeneration. It is a rare disease that is found in all populations, but it is particularly prevalent in Ashkenazi Jews of Eastern European origin. There is no treatment, but research aimed at treating the disease by blocking synthesis of the affected molecules has been ongoing since the late 1990s. Carriers can be identified by DNA or **enzyme** tests and prenatal diagnosis is available to at-risk families.

History and Disease Description

In 1881 Warren Tay, a British ophthalmologist, observed a “cherry red spot” in the **retina** of a one-year-old child with mental and physical retardation. Later, in 1896 Bernard Sachs, an American neurologist, observed extreme swelling of neurons in autopsy tissue from affected children. He also noted that the disease seemed to run in families of Jewish origin. Both physicians were describing the same disease, but it was not until the 1930s that the material causing the cherry-red spot and neuronal swelling was identified as a ganglioside lipid and the disease could be recognized as an “inborn error of metabolism.” The term “ganglioside” was coined because of the high abundance of the brain lipid in normal ganglion cells (a type of brain cell). In the 1960s, the structure of the Tay-Sachs ganglioside was identified and given the name “ G_{M2} ganglioside” (Figure 1).

Gangliosides are glycolipids. The lipid component, called ceramide, sits in the membranes of cells. Attached to it and sticking out into the extra-



cellular space is a linked series of different sugars, the “glyco” portion of glycolipid. The basic function of gangliosides is not well understood, but they appear to have roles in biological processes as diverse as cell-to-cell recognition, differentiation, and in the repair of damaged neurons.

Gangliosides, like most cell components, are broken down and regenerated as part of normal cellular metabolism. The breakdown or “catabolism” of gangliosides occurs in the lysosome, a specialized vesicle that is analogous to the vacuole of plants. In the lysosome a series of acid hydrolases (degradative enzymes) removes each sugar, one at a time, until the ceramide lipid is all that remains. In Tay-Sachs disease, one of the lysosomal hydrolases, Hex A, is defective or completely absent, so the degradative process is blocked before completion. The result is the accumulation of G_{M2} ganglioside, the last molecule before the Hex A block in the catabolic sequence.

Since breakdown is blocked while synthesis continues, the result is a progressive accumulation of GM2 ganglioside and massive swelling of the lysosomes and hence of the neurons containing them. This is the basis of

Figure 1. Molecular basis for Tay-Sachs disease. Absence of or defect in the Hex A protein prevents complete processing of GM2 ganglioside.



Figure 2. The macular cherry-red spot on the eye is an indicator of Tay-Sachs disease, an autosomal recessive disease that affects the nervous system.

polypeptide chain of amino acids

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

endoplasmic reticulum network of membranes within the cell

Golgi network system in the cell for modifying, sorting, and delivering proteins

the neuron swelling observed by Sachs and the cherry-red spot described by Tay. The cherry-red spot is due to the white appearance of swollen neurons of the retina surrounding the normally red fovea centralis (central depression in retina and site of maximum vision acuity) in the back of the eye (Figure 2).

Newborns with Tay-Sachs disease appear normal at birth. By six months of age, parents begin to notice that their infant is becoming less alert and is less responsive to stimuli. The affected infant soon begins to regress and shows increasing weakness, poor head control, and inability to crawl or sit. The disease continues to progress rapidly through the first years of life, with seizures and increasing paralysis. The child eventually progresses to a completely unresponsive vegetative state. Death is often caused by pneumonia because of the child's weakened state. Some forms of Tay-Sachs disease are much milder with onset of the disease later in childhood or even adulthood. We now know that these forms of the disease are caused by less severe mutations in the *HEXA* gene.

Molecular Biology: Understanding Tay-Sachs Disease

Hex A is composed of two **polypeptide** subunits, one called α and one called β (Figure 1). One other form of the enzyme, Hex B, is composed of two β subunits. In Tay-Sachs disease, it is the α subunit that is mutated so that patients have a defective Hex A, while Hex B remains unaffected. However, Hex B is not active toward G_{M2} ganglioside and can not substitute for Hex A. Some patients have a disease similar to Tay-Sachs, with the absence of both Hex A and Hex B. This condition, now called Sandhoff disease, was first described by Konrad Sandhoff in the 1960s and is due to mutations in the β subunit.

One more protein is involved in the disease. It is called the G_{M2} activator and is essential to the breakdown of G_{M2} ganglioside. The protein forms a complex with the G_{M2} ganglioside, converting the G_{M2} from a hydrophobic, membrane-liking molecule to one that is hydrophilic (water loving) so that it can be successfully hydrolyzed by Hex A in the lysosome. Mutations in the G_{M2} activator gene, called *GM2A*, can also cause a Tay-Sachs-like disease, although it is exceedingly rare.

In sum, mutations in any of three genes can cause the disease: *HEXA*, *HEXB*, or *GM2A*. As a group, patients with any of these diseases are said to have $GM2$ gangliosidosis. Tay-Sachs disease refers specifically to the most common form of the disease, caused by mutations in the *HEXA* gene.

The *HEXA* gene is one of about 30,000 to 70,000 genes in the human genome. It is of average size at about 35,000 **base pairs** in length and contains fourteen exons. The remainder of the gene is made up of thirteen introns that separate the exons from one another. Extensive research has given us a clear picture how the enzyme is synthesized, processed through the **endoplasmic reticulum** and **Golgi network** of the cell, and sent to the lysosome. This understanding of the cell biology of Hex A has had an important impact on our understanding of mutations in Tay-Sachs disease. Some affect enzyme function, that is, they occur near the “active” site of the enzyme and block its activity, while others affect the biosynthetic processing of the protein. The latter type of mutations may not affect enzyme activity at all, but causes disease because the enzyme fails to reach the lysosome to carry out its biological role.

Mutations and Founder Effect

To date, nearly 100 mutations have been identified in Tay-Sachs disease. Ashkenazi Jews have two common mutations that cause the severe, infantile form of the disease. One, accounting for 80 percent of mutant **alleles** carried in the population, is the loss of four nucleotides in exon 11 of the gene. This causes a “frameshift” in the reading of the genetic code and the inability to generate a complete protein. The second mutation, accounting for about 15 to 18 percent of mutations, is a splice junction mutation, a defect in processing nuclear RNA to form the mature messenger RNA that makes its way to the cytoplasm to direct protein synthesis. When this mutation is present, splicing of intron 12 fails to occur properly, and a functional protein fails to be synthesized. In both cases, the result is the absence of the α subunit and hence Hex A.

Ashkenazi Jews and other populations also have mutations that cause amino acid substitutions. One such mutation, accounting for about 3 percent of mutations in Ashkenazi Jews, results in a glycine to serine substitution in exon 7. This mutant α subunit is synthesized and an abnormal Hex A is produced. It is sufficiently active so that patients with this mutation as one of their two mutant alleles have sufficient “residual” Hex A activity to produce a mild, adult form of the disease.

This finding of three predominant mutations causing Ashkenazi Jewish Tay-Sachs disease was unexpected. Most medical scientists thought a single mutation would have acted as a “founder” mutation and over time increased in frequency to the level at which it is found today. It is believed that the first Tay-Sachs mutation may have entered the population about 1000 years ago. It increased in frequency either by random **genetic drift** or possibly through selection for presence of the gene in heterozygous carriers. This latter interpretation is controversial, but it has been suggested that carriers might have been more resistant to tuberculosis than normal individuals so that they had a greater chance of surviving the epidemics of centuries ago, thereby resulting in a steady increase in the frequency of the mutant allele in Ashkenazi Jews. Another group with a founder mutation, a large deletion of the 5' (“five prime,” or front) end of the gene, are French Canadians from the Lac Saint-Jean region of Quebec.

Carrier Testing and Prenatal Diagnosis

One in thirty Ashkenazi Jews is a carrier of one of the Tay-Sachs mutations. This is about ten times the frequency of carriers in non-Jews. Until 1970 it is estimated that about one in 4,000 births among Ashkenazi Jews was of a Tay-Sachs baby. This produced a great desire to develop a carrier and prenatal test shortly after the enzyme defect was identified. Michael Kaback spearheaded a carrier testing program that, by 2000, had tested well over 1 million Ashkenazi Jews, mainly in North America and Israel. This led to a drop in the incidence of Tay-Sachs disease to less than one-tenth of its previous level.

The testing program has been so successful because it is organized through Jewish community groups, with the active participation of geneticists who conduct the tests. During the 1970s and 1980s the test measured the level of Hex A activity in serum or white blood cells. With the identification of the

alleles particular forms of genes

genetic drift evolutionary mechanism, involving random change in gene frequencies

